

MDP-GRL: Multi-disease Prediction by Graph-enabled Representation Learning

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Abstract—In recent years, automatic disease prediction based on electronic health records (EHRs) has emerged as a focal area of research in medical informatics. While successfully facilitating disease diagnosis, this technique still suffers from many limitations caused by the complexity of medical data, particularly the diverse relations and shared risk factors among multiple diseases. Besides, the data sparsity and imbalanced problem in EHR also undermines the effectiveness of existing approaches. Therefore, new approaches are urgently needed to accommodate the EHR features better and make effective predictions on individuals' potential diseases. To address the above challenges, this paper proposes MDP-GRL, a novel multi-label disease prediction model based on graph-enabled representation learning. Specifically, MDP-GRL constructs a medical knowledge graph (MKG) based on the patient and disease information in EHR and then employs a graph neural network (GNN) to realise the disease prediction. To address the data sparsity issue, it incorporates supplementary data for both patients and diseases, i.e., enriching patient nodes by personal basic information, examination indicators, and illness history, and supplementing disease information with comorbidity information, prevalent populations, common causes, and diagnostic basis. To mitigate the data complexity issue, MDP-GRL considers four different relation patterns in MKG, which optimizes the modelling capabilities. To address the data imbalance problem, it introduces an attention mechanism and self-adversarial negative sampling strategy, which further enhance MDP-GRL's ability to identify error-prone and minority samples. Comprehensive experiments and ablation studies are conducted based on the MIMIC-IV dataset. The results demonstrate MDP-GRL's superiority in multi-disease prediction compared with state-of-the-art approaches.

Index Terms—Multi-disease prediction, medical knowledge graph, graph neural network, data imbalance, data sparsity

APPENDIX

To further investigate the impact of different modules in MDP-GRL, we conduct a set of ablation studies as follows. Please note that PyRec has the best performance across the five representative approaches in common scenarios where $x \in 1$, e.g., P@3, P@5, etc. Therefore, if necessary, we compare MDP-GRL against only PyRec in the ablation study.

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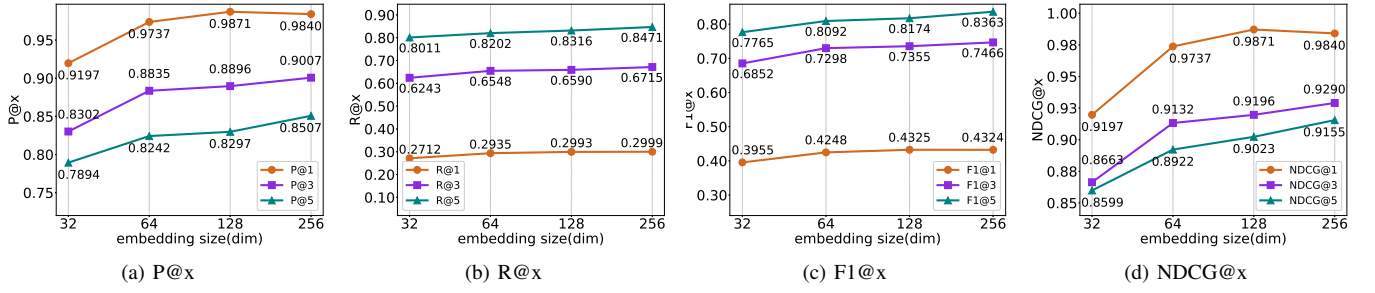
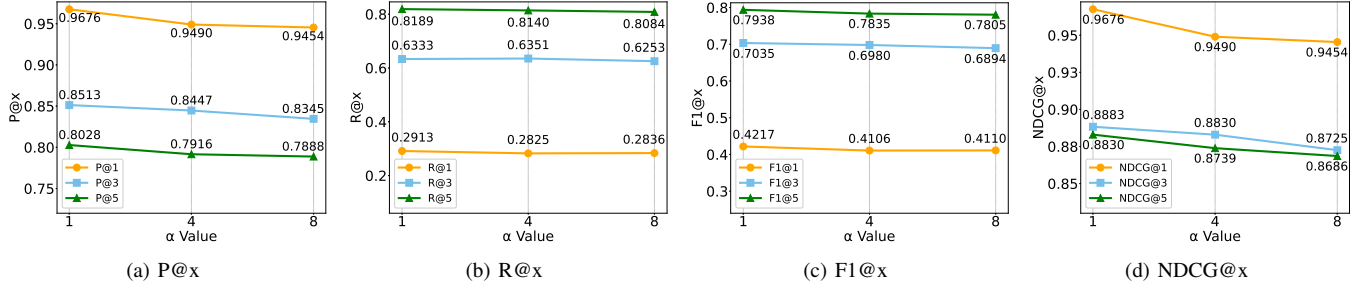
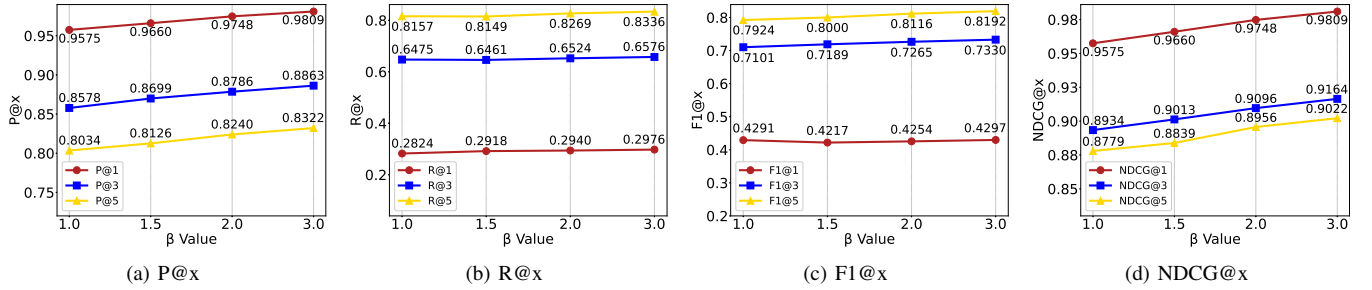
1) Impact of Embedding Size dim : MDP-GRL employs GNN to embed those MKG nodes to a complex latent space, and then updates those embeddings by distilling useful information from their neighbour nodes' embeddings. The embedding size dim determines the size of each node embedding. To investigate the impact of embedding size on MDP-GRL's performance, we vary dim to $\{32, 64, 128, 256\}$. The experiment results are shown in Fig. 1.

Briefly, when the embedding size dim increases from 32 to 128, MDP-GRL's performance improves by 6.54%, 6.57%, 7.32%, and 6.14% in Precision, Recall, F1, and NDCG metrics, respectively, on average across all cases. This indicates that a higher embedding size allows the model to more effectively present a piece of information in the knowledge graph to the latent space, thereby enhancing its embedding capacity and prediction accuracy. However, when the embedding size increases from 128 to 256, the model's performance experiences minor improvement. For example, the performance changes in the Precision, Recall, F1, and NDCG metrics are less than 1.5%. Specifically, they are 1.15%, 1.32%, 1.27%, and 0.72%, respectively. The performance even slightly declines in P@1, R@1, F1@1, and NDCG@1. This phenomenon shows that performance tends to plateau at this point, i.e., once the embedding size reaches a certain level, the model has already fully learned the complex information between different nodes and relationships through GNN. In this case, further increasing the embedding size has a limited effect on performance improvement but will inevitably introduce excessive time consumption during the training.

2) Impact of Temperature Parameter α : The temperature parameter α determines how MDP-GRL select negative samples, as introduced in Eq. (4). To investigate the impact of the α on MDP-GRL's performance, we set $\alpha \in \{1, 4, 8\}$. The results are shown in Fig. 2.

We can find that when α increases from 1 to 8, the four performance metrics including Precision, Recall, F1, and NDCG decrease by an average of 2.00%, 1.73%, 2.07%, and 1.90% respectively, on average across all cases. Furthermore, when α increases from 1 to 4, the decline in performance is relatively rapid. This is because when α is small, the probability distribution of negative samples becomes steeper. This means MDP-GRL tends to select negative samples with higher scores for training, i.e., those samples that the MDP-GRL deems less likely to be true ternaries. This helps the model pay more attention to those small sample data that might otherwise be overlooked, addressing the issue of data imbalance to some extent. However, when α increases from 4 to 8, the decline in model performance gradually flattens out. This is because as α further increases, the probability distribution of negative samples becomes more flat, and the probability of each negative sample being selected becomes more similar. Even as the α value continues to increase, changes in model performance will tend to flatten out.

3) Impact of β : The parameter β regulates the sensitivity of the *softplus* function to inputs in Eq. (6). To study its impact on MDP-GRL's performance, we set four values, i.e., $\beta \in \{1, 1.5, 2, 3\}$. The results in Fig. 3 show that the four metrics Precision, Recall, F1, and NDCG are improved by an average of 3.12%, 3.04%, 2.25%, and 2.60%, respectively, when β increases. This is attributed to the characteristics of the *softplus* function and its interaction with the

Fig. 1: Impact of Embedding Size dim Fig. 2: Impact of Temperature Parameter α Fig. 3: Impact of β

β parameter. Specifically, the *softplus* function exhibits a particular characteristic when processing negative inputs: when β increases, the process by which the function value approaches 0 becomes steeper. In the experimental setup of this study, negative sample scores, when subtracted from γ ($\gamma = 3$ in this experiment), typically result in negative numbers. Therefore, as β increases, the response of the *softplus* function to negative inputs becomes more sensitive, and the model can distinguish between positive and negative samples more accurately, thus it improves MDP-GRL's performance.